2025 BATCH

PRINCIPLES OF DATA SCIENCE

CW 2 (PORTFOLIO) -7144

Agentic AI Workflow for Automated Cryptocurrency Trading

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COMSCDS252P-008

COVENTRY UNIVERSITY
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1. Introduction

In financial markets, particularly cryptocurrency markets, time to time generates a vast volume of dynamic and unstable data. It makes a high challenge for analyzing manually. Traders are required to continuously monitor price movements, technical indicators, and external information such as news, blogs, and posts for making accurate decisions. This complexity makes for delayed reactions, incompatible decision-making, and enhanced exposure to risk. Therefore, there is a requirement from the traders for automated system that have ability to assist systematic trading strategies by using multiple resource of information.

Agentic Artificial Intelligence (AI) workflow provide the solution by combining automated modules that perform specific analytic tasks. These workflows collect data, clean & preprocess data, analyze patterns of data, gather external information, generate decisions and manage risk in a coordinated and insightful way. In financial markets, they execute trading strategies using data instead of personal feelings for improve consistency, reliability, and scalability.

This project designs and implements an agentic AI-based trading workflow that focuses on cryptocurrency markets. The system integrates historical market data and related news information to support automated Sell, Buy or Hold decision making. Market analysis is led using statistical techniques and technical indicators while external information is incorporated through a dedicated information retrieval module that processes cryptocurrency-related news headlines. Furthermore, the system includes a decision engine, a risk management module, backtesting component to evaluate performance, a Large Language Model (LLM) to do analytical interpretation using generated results, and send a notification including essential reports, figures, and an interpretation report as an email to a specified email address.

In addition to analytical ability, the system comprises cloud-based infrastructure using Amazon Web Services (AWS), EC2 instance performs computation, Amazon S3 stores raw and processed data, and Amazon RDS PostgreSQL database maintains structured outputs. The system uses the `openai/gpt-oss-120b` LLM to generate an analytical interpretation of the results.

Overall, the system provides a comprehensive framework for analyzing and evaluating the trading strategies.

1.1. Git Repository Access

Public Git Repository:

https://github.com/JanakaUPerera/agentic_ai_trading_comscds252p008.git

2. Problem Definition

2.1. Background

The cryptocurrency markets denote high instability, rapid price fluctuations, and continuous trading activity across the global markets. Unlike traditional markets, cryptocurrency markets operates 24/7 hours, and wide range of factors such as technical indicators, investor sentiment, regulatory developments, and security-related events, influence them to operate as such. This creates a complex environment for traders when they interpret and manage large amount of different type of information in real time.

Manual trading in such an environment, it is difficult to make accurate and reliable decisions continuously and quickly while monitoring technical indicators and staying updated with the latest market news simultaneously. Furthermore, traders make decisions based on personal feeling instead of data, causing a tendency toward bias and reduce reliability.

2.2. Problem Statement

The primary problem addressed in the project is the difficulty of making consistence and data driven decisions on high unstable cryptocurrency markets. Traders must combine multiple sources if information, including historical price data and external market news events to identify profitable opportunities while managing risk effectively.

Traditional approaches tend to have an incomplete decision-making framework by technically analyzing and news analyzing separately. Therefore, traders need integrated system that combine market analysis, external information retrieval, and risk management into a unified framework.

2.3. Objectives

The main objective of the project is to design and implement an Agentic AI-based trading workflow with capability of autonomously analyzing market conditions and generating trading decision.

The specific objectives are;

- ✓ Collecting cryptocurrency data using reliable API and processing them.
- ✓ Performing Exploratory Data Analysis (EDA) to identify patterns, volatility, and relationships between assets.

- ✓ Engineering technical indicators such as Moving Averages, Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD) for market analysis.
- ✓ Retrieving and analyzing cryptocurrency –related news, blogs, and posts from external resources.
- ✓ Designing a decision engine that generates Buy, Sell, or Hold signals bases on combined inputs.
- ✓ Incorporating the risk management module that reduces exposure during high-risk conditions.
- ✓ Evaluating the performance of the trading strategy using backtesting techniques.
- ✓ Implementing cloud-based storage and computation using AWS services.
- ✓ Enhancing interpretability through an LLM-based analytical interpretation module.

3. Data Collection

The system collects historical cryptocurrency market data by using the Yahoo Finance API through the `yfinance` Python library [1]. The dataset includes daily observations for ten cryptocurrency assets:

- ✓ BTC-USD – Bitcoin USD
- ✓ ETH-USD – Ethereum USD
- ✓ BNB-USD – Binance Coin USD
- ✓ SOL-USD – Solana USD
- ✓ XRP-USD – XRP USD
- ✓ ADA-USD – Cardano USD
- ✓ DOGE-USD – Dogecoin USD
- ✓ TRX-USD – TRON USD
- ✓ AVAX-USD – Avalanche USD
- ✓ LINK-USD – Chainlink USD

(Appendix A.1.1)

*Note: "**-USD**" denotes the fiat currency pairing, representing the price of one unit of the cryptocurrency in United States Dollars.*

The reason for choosing these assets is to provide diverse representation in terms of market capitalization, technology, and use case. Bitcoin and Ethereum represent the largest and most established assets, and Binance, Solana, and Avalanche reflect high-growth platforms with strong ecosystem activity. XRP, Cardano, and Tron offer variations in network design and adoption. Dogecoin represents a highly sentiment-driven asset, and Chainlink adds exposure to decentralized oracle infrastructure. This diversity integrates different market behaviors, volatility patterns, and responses to external factors into analysis for making the trading workflow more robust and comprehensive.

For each asset in the dataset contains Open, High, Low, Close, and Volume (OHLCV) values over the selected time period. This data establishes the foundation for performing market analysis, computing technical indicators, and generating trading signals.

The system collects data 3 years back from the current date. It allows capturing multiple market regimes, including bullish trends, bearish corrections, and high-volatility phases. It does not collect too-old data to ignore collecting less relevant data.

4. Data Preprocessing

4.1. Data Cleaning

This includes removing duplicate records, verifying data types, ensuring all numerical values are correctly formatted, and convert date column to datetime data type. Negative price values of columns such as open, high, low, close, and adjusted close are considered invalid data, and the system removes their rows from the dataset. Eventually the system saves the processed dataset into new CSV file sorting by date and ticker values. (Appendix A.1.2.1)

4.2. Handling Missing Values

The missing values of columns such as open, high, low, close, and adjusted close are filled using forward-filling techniques. The system replaces zero for empty values of the volume column and removes rows with excessive missing information, such as date and ticker columns, from the dataset.

5. Exploratory Data Analysis (EDA)

5.1. Descriptive Statistic

The system computes descriptive statistics to summarize the key characteristics of dataset. It analyses metrics such as mean return, standard deviation, minimum values, maximum values, and median to understand the overall behavior of each cryptocurrency assets. (Appendix A.2.2.1)

Such as BTC and ETH high-value assets exhibit large standard deviations, reflecting significant price fluctuation and higher market risk. Conversely, such as TRX and ADA show comparatively small variation, indicating low market risk and stable behavior. The assets with higher trading volumes are more stable and easier to trade because more buyers and sellers are available and vise-versa.

5.2. Price Trends

The system uses the time-series plot to visualize price movements over the chosen period. These plots reveal trends, cycles, and sudden price changes by highlighting the inconsistence nature of the cryptocurrency. (Reactive normalized price trends provides in Appendix A.2.1.3)

Furthermore, the system generates the normalized price trending plot for fair comparison of performance because each assets' prices are on completely different scales. Normalization provides each line starts from the same point and differences show as growth/decline. Initially the each lines start from 100.

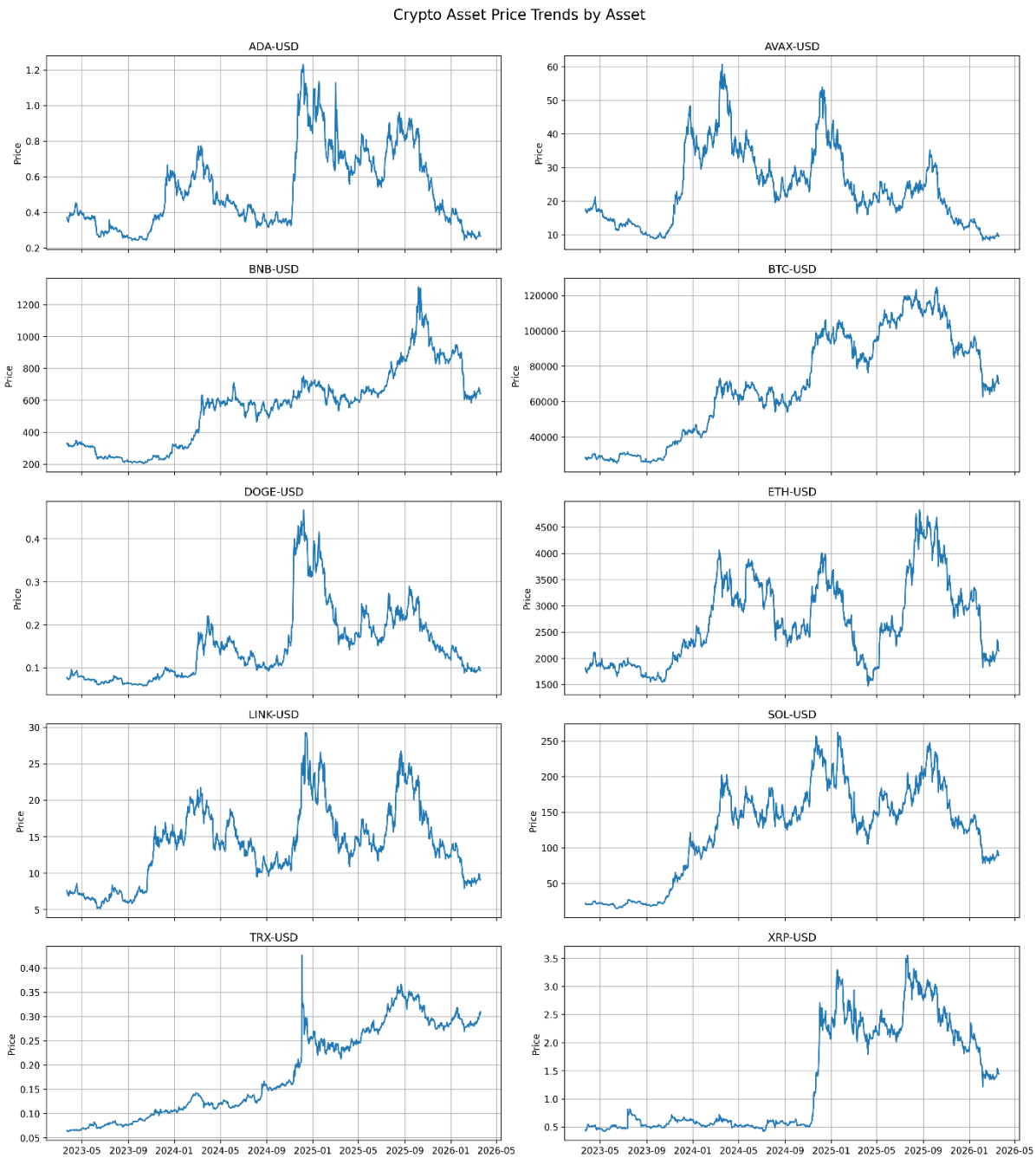

Figure 5.2.1: Price Trends Sub Plots

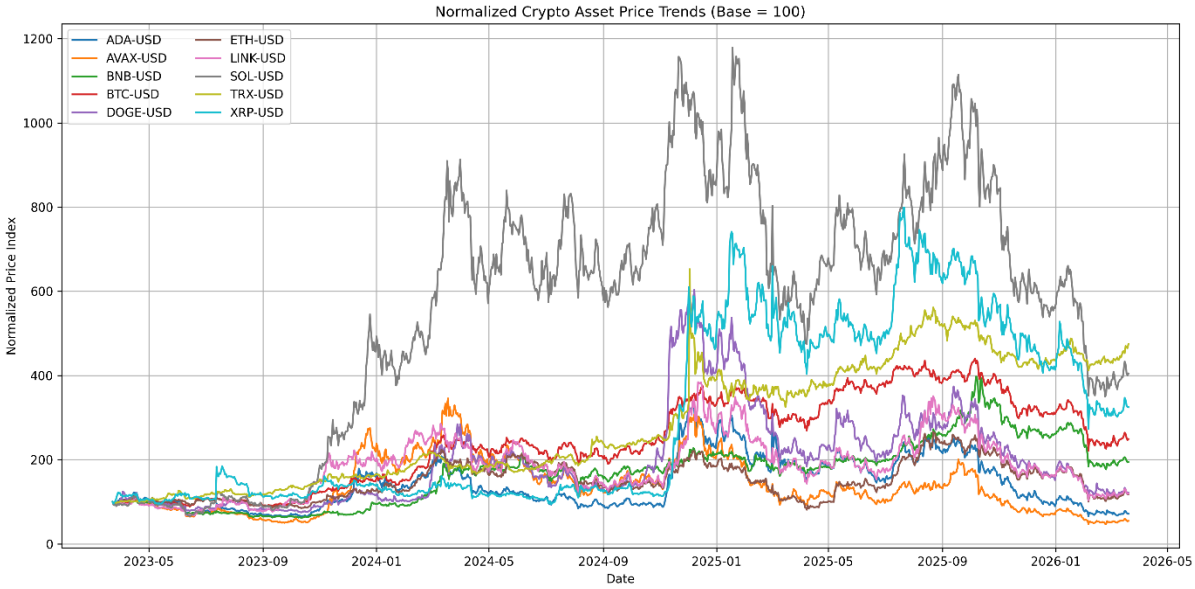

Figure 5.2.2: Normalized Price Trends

The normalized price trends represent that most cryptocurrencies follow similar market cycles by rapidly growing and then correcting, which means common factors such as news, regulations, and global economy effect on the whole market. Coins like SOL, XRP, and AVAX change more dynamically. It means, the market momentum and publicity effects on them. Meanwhile, larger assets such as BTC and ETH move more smoothly, showing they are more stable and liquid.

5.3. Volatile Analysis

This measures the variability of asset returns over time. It helps identify high unstable and risk periods, which are critical for trading and making risk management decisions. (Reactive rolling volatility provides in Appendix A.2.1.4)

Figure 5.3.1: Rolling Volatility by Asset

Coins like SOL, XRP, and TRX have bigger and sudden changes, which means they are riskier and react more to market shocks. But such as BTC and ETH are more stable, showing they are consistence. Overall many coins become volatile at the same time which means market events effect on all of them together.

5.4. Correlation Analysis

Correlations analysis performs to identify the relationship between each cryptocurrency assets. (Reactive correlation heatmap provides in Appendix A.2.1.1, Appendix A.2.2.4)

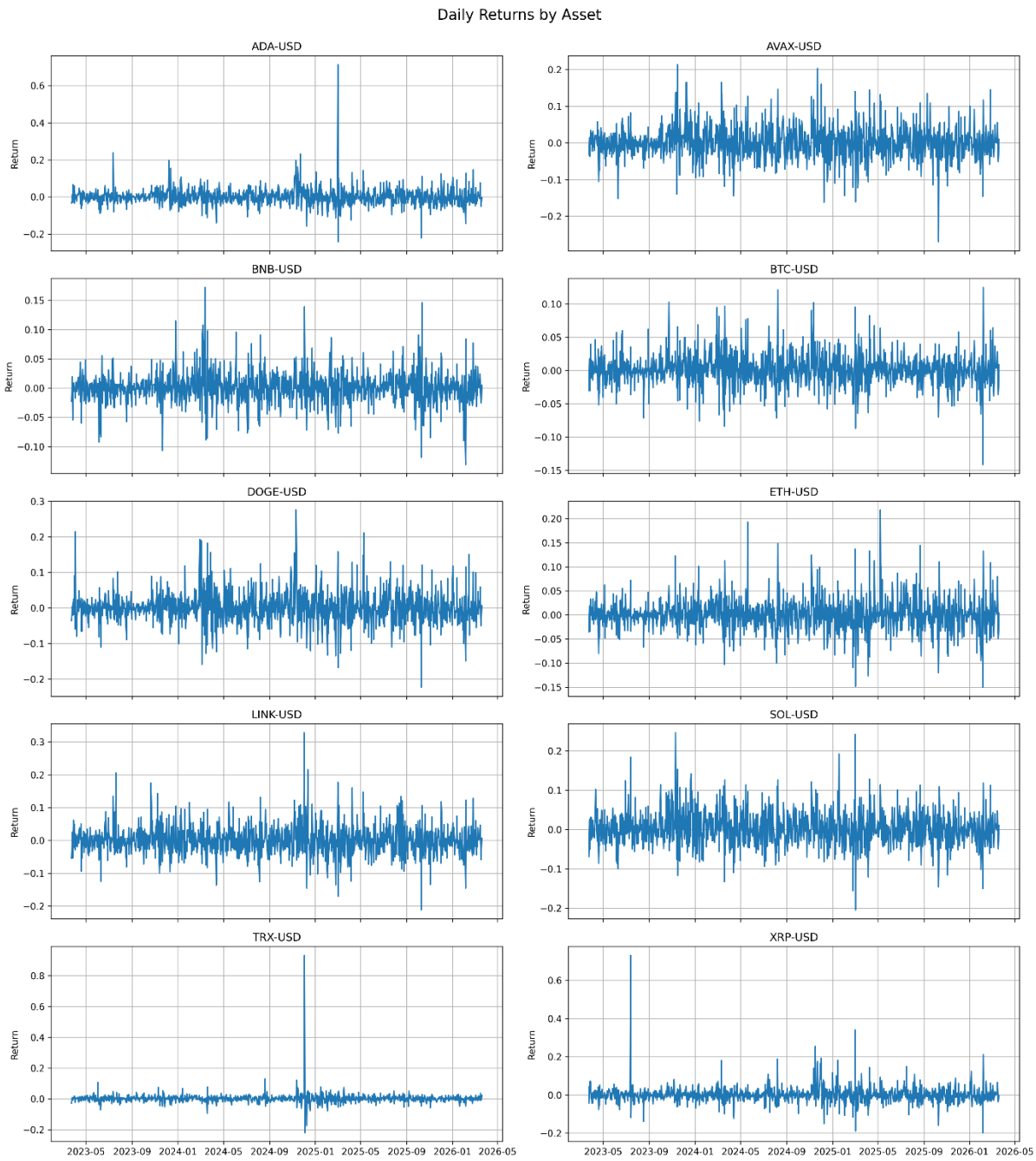

Figure 5.4.1: Daily Returns by Asset

Crypto prices change frequently but usually stay around small profits and losses. Coins like SOL, AVAX, and DOGE have large and more frequent changes, which means it has higher short term risk but BTC and ETH are more stable. Meanwhile, TRX and XRP sometimes perform extreme spikes, which means they are unpredictable.

The correlation heatmap is generated using these daily returns.

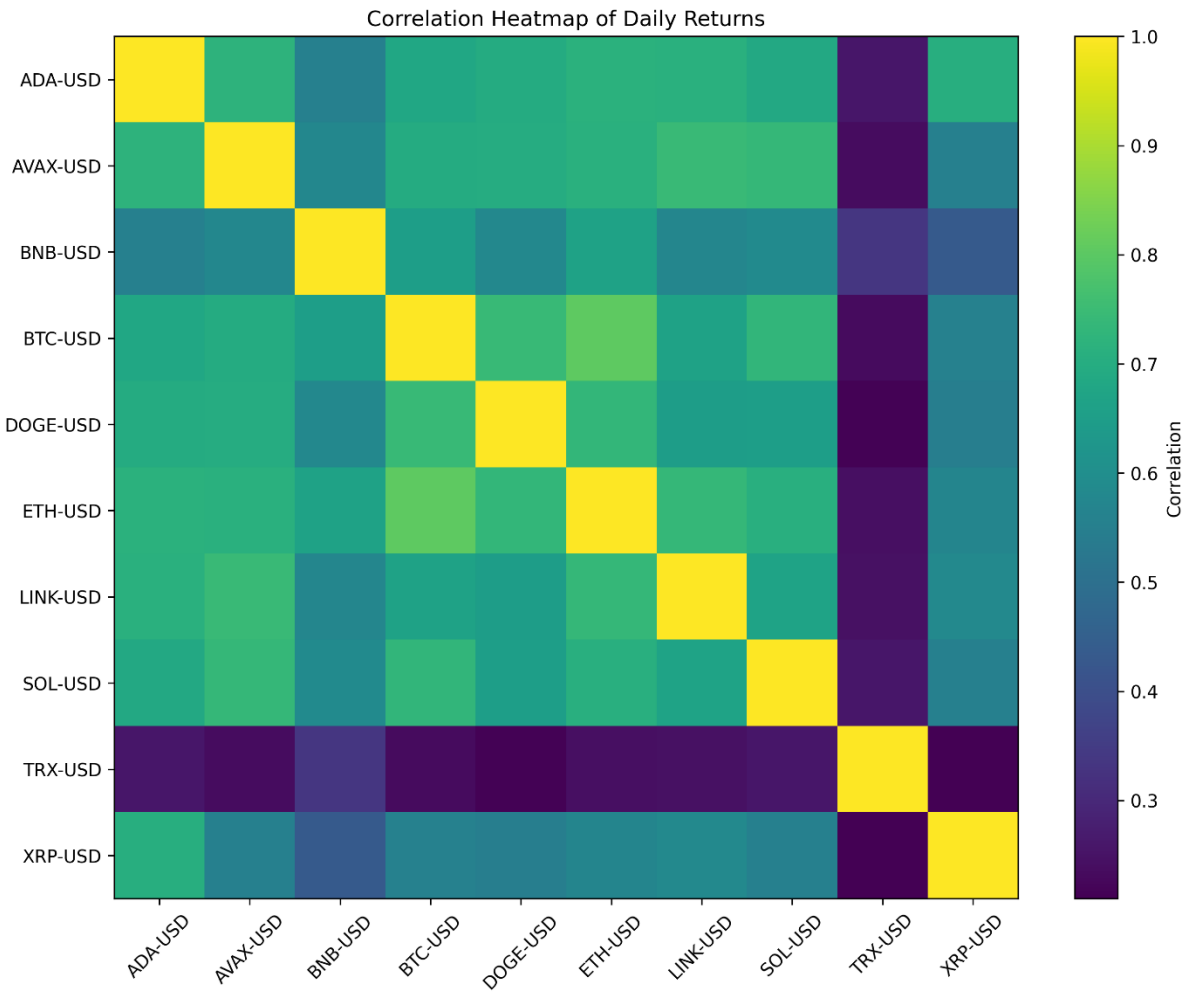

Figure 5.4.2: Correlation Heatmap of Daily Returns

The heatmap indicates, most cryptocurrency move in the same direction under similar market conditions. BTC and ETH strongly influence the overall market. The TRX moves quite differently, which means traders can't predict TRX, inspecting overall market behavior.

6. Feature Engineering

6.1. Technical Indicators

The system computes several technical indicators to capture market trends and momentum. Moving average assists to identify short-term and long-term trends and RSI helps to detect overbought and oversold conditions. The system uses MACD to analyze momentum and trend direction. (Appendix A.1.2.4)

6.2. Signal Generation

The system generates rule-based trading signals based on computed indicators. RSI values below the threshold indicates a potential for buying opportunities and above the threshold suggests sell condition. MACD signals were used to confirm trend direction. The system finally combines all of these signals to produce Buy, Sell, and Hold recommendation for each assets. (Appendix A.1.2.5)

7. Clustering

The clustering applies K-Means clustering to group cryptocurrency assets based on their statistical and technical characteristics. Features such as averages, volatility, RSI, and MACD helps to identify clusters of assets containing similar behaviors. (Appendix A.1.2.2, Appendix A.2.2.2, Appendix A.2.2.3)

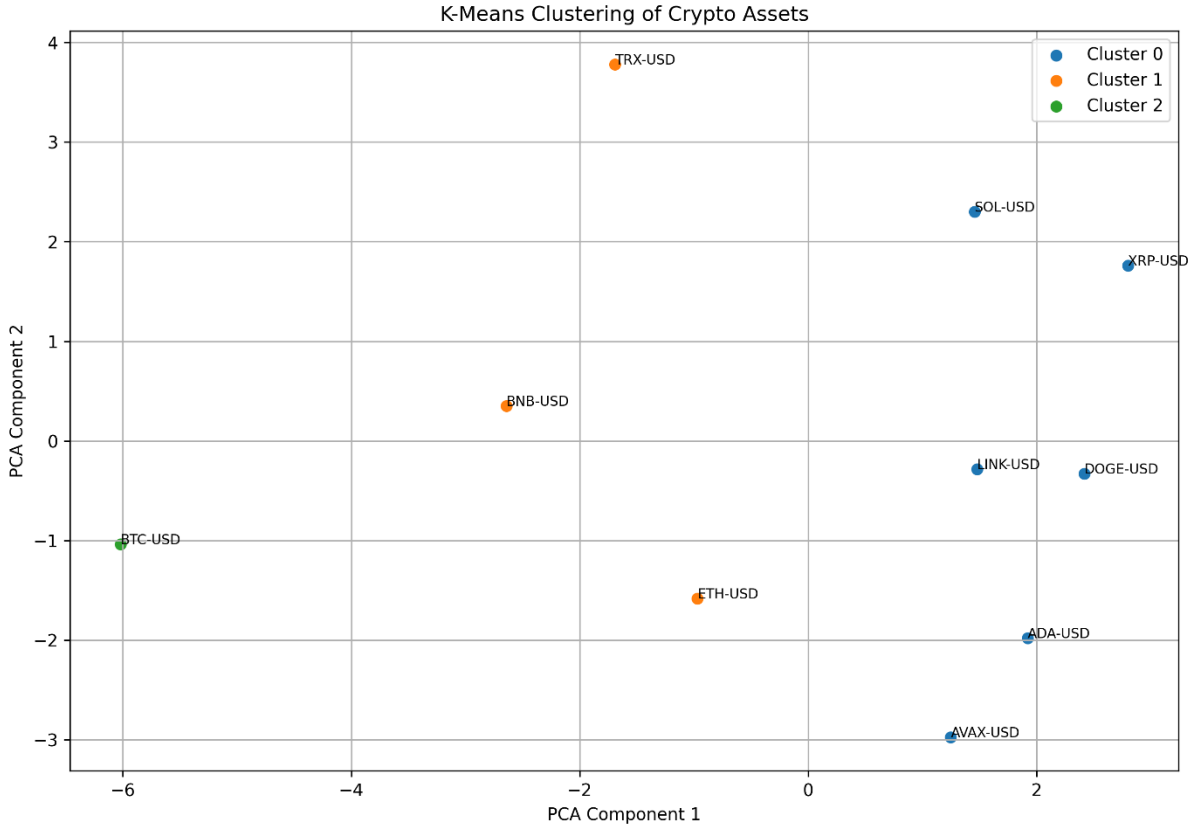

Figure 7.1: K-Means Clustering of Crypto Assets

Coins like SOL, XRP, AVAX, ADA, DOGE, and LINK behaves similarly. These coins are more volatile and react quickly to market sentiment, often showing high growth but also high risk. The system identify this as a one cluster.

BTC plays an individual role here; it behaves differently from other cryptocurrencies. BTC is more stable and has higher liquidity, making it dominant in the crypto market.

Coins like ETH, BNB, and TRX are not as volatile as the first cluster but not as stable as BTC.

Therefore, clustering identifies clear three clusters.

8. Information Retrieval Module

8.1. RSS News Collection

The system collects cryptocurrency-related news from multiple online sources using Really Simple Syndication (RSS) feeds. The selected sources includes;

- ✓ CoinTelegraph
- ✓ CoinDesk
- ✓ Decrypt
- ✓ CryptoNews
- ✓ NewsBTC
- ✓ Bitcoin Magazine
- ✓ The Block
- ✓ AMBCrypto

(Appendix A.1.3.1)

The `feedparser` python library parses the RSS feeds and extracts headlines along with metadata such as publication time and source [2]. The system uses these headlines to capture external market sentiment and identify events that may influence price movements.

8.2. Sentiment Classification

The system categorizes news headlines mainly into five classes by analyzing using rule-based sentiment classification approach.

- ✓ Bullish
- ✓ Bearish
- ✓ Regulatory
- ✓ Security
- ✓ Neutral

The system checks headlines for specific keywords to identify the sentiment. The system has predefined keywords for each sentiment and each asset; therefore, the system easily decides which news indicate which sentiment for which asset. (Appendix A.1.3.2)

8.3. Weighted Scoring Mechanism

The system follows the weighted scoring mechanism to improve the quality of sentiment analysis. The final sentiment score of each headlines is powered by;

- ✓ Source Weight: Scoring for the credibility and reliability of different news sources
- ✓ Recency Weight: Giving higher priority and importance to more recent news
- ✓ Confidence Weight: Reflecting the strength of keyword matches in the classification process

The system combines all these weights for calculating the final weight at the asset level to generate an overall sentiment signal. (Appendix A.1.3.3)

9. Decision Engine

9.1. Input Signals

The decision engine integrates multiple sources of information to generate trading decisions. The system uses these inputs as primary inputs;

- ✓ Technical signals derived from indicators such as RSI, MACD, and moving average
- ✓ News-based signals including sentiment category, market impact, and asset-level bias
- ✓ Weighted directional news scores reflecting the strength of sentiment

9.2. Decision Logic

The decision engine uses combined technical signals and news sentiment to decide what action should be taken from Buy, Sell, or Hold. Technical signals decide the main direction whether it's Buy, Sell, or Hold. Meanwhile, news sentiment adjust the signals according to positive, negative, and neutral news.

9.3. Conflict Resolution

In case where technical and news signals conflict, the system applies predefined rules to solve inconsistency. For example, if technical indicators suggest a Buy signal but related strong negative news is present, the system may downgrade the decision to hold. Similarly, security-related news events can override other signals and trigger a Sell decision to reduce risk exposure.

(Appendix A.1.3.3, Appendix A.1.3.4, Appendix A.1.3.5)

10. Risk Management

10.1. Volatility Risk

The system assess volatility risk using rolling volatility measures. It categorizes assets into low, medium, or high volatility levels based on their distribution. High volatility-risk conditions indicate uncertainty and potential risk.

10.2. News-Based Risk

The risk management module evaluates the news-driven risk using the sentiment classification method. The method treats security-related news such as hacks and breaches as severe risks and negative regulatory developments as high risk. If news bias is strongly negative, the method treats it as high bias, and if it's negative, it treats it as medium. Otherwise as low.

10.3. Position Risk

The module derives position risk from technical indicators such as RSI, volatility, recent returns and decision scores. Extreme RSI values or sudden large price movements indicate potential overbought or oversold conditions, increasing the likelihood of reversals. Such conditions were classified as high position risk.

10.4. Risk Overrides

The risk management module applies overrides to the decisions generated by the decision engine. For example;

- ✓ High volatility may downgrade a Buy decision to Hold
- ✓ Severe security risk may trigger a Sell decision
- ✓ Negative news conditions may prevent new positions from being opened

These overrides ensure that trading decisions are aligned with risk conditions, improving the robustness of the strategy.

11. Backtesting

11.1. Strategy Setup

The system evaluates the trading strategy using historical backtesting. It converts the risk-adjusted trading decisions into positions, where Buy corresponds to long position which means expect the increasing price, Sell to a short position which means expect the decreasing the price, and Hold to no position which means no changes in price. Furthermore, it includes scoring method to adjust size of the positions according to strength of the decision score. This approach allows the strategy to adjust exposure based on confidence, rather than using fixed position size.

11.2. Performance Metrics

The system evaluates the backtesting module's performance of the strategy, using several metrics.

- ✓ Cumulative Return: Measures total growth of the portfolio
- ✓ Benchmark Return: Compares performance against average market returns
- ✓ Maximum Drawdown: Measures the largest loss from peak to trough
- ✓ Sharpe Ratio: Evaluates risk-adjusted returns
- ✓ Win Rate: Percentage of profitable trading periods
- ✓ Trade Count: Total number of executed trades

  Transaction cost is also incorporated to provide a more realistic evaluation of strategy performance.

11.3. Results and Analysis

Note: Results and analysis could vary according to day-to-day cryptocurrency data. This result based on data that were gathered at 2026-03-26.

The backtesting results demonstrate that the strategy is capable of capturing market trends while managing risk through the integrated risk management module. The combination of technical indicators and news-based signals improves decision quality, particularly during periods of strong market sentiment.

The strategy shows competitive performance compared to the benchmark, with controlled drawdowns and consistent trading behaviors.

Comparative Performance Metrics: Agentic AI Strategy vs. Benchmark

Performance Metric	Strategy	Benchmark (Equal-Weight)
Final Portfolio Growth ($1 initial)	**3.538 (253.8%)**	2.469 (146.9%)
Total Return	**253.8%**	146.9%
Maximum Drawdown	**-29.2%**	-60.2%
Sharpe Ratio	**1.259**	0.667
Average Daily Return	**0.1287%**	0.1321%
Total Trades Executed	**1,145**	N/A
Active Position Days	**4,050**	N/A
Win Rate (Daily)	**47.3%**	N/A
Winning / Losing Days	**445 / 495**	N/A

Figure 11.3.1: Comparative Performance Metrics: Agentic AI Strategy vs. Benchmark

(Appendix A.1.3.6, Appendix A.2.2.5)

11.3.1. Portfolio Growth Analysis

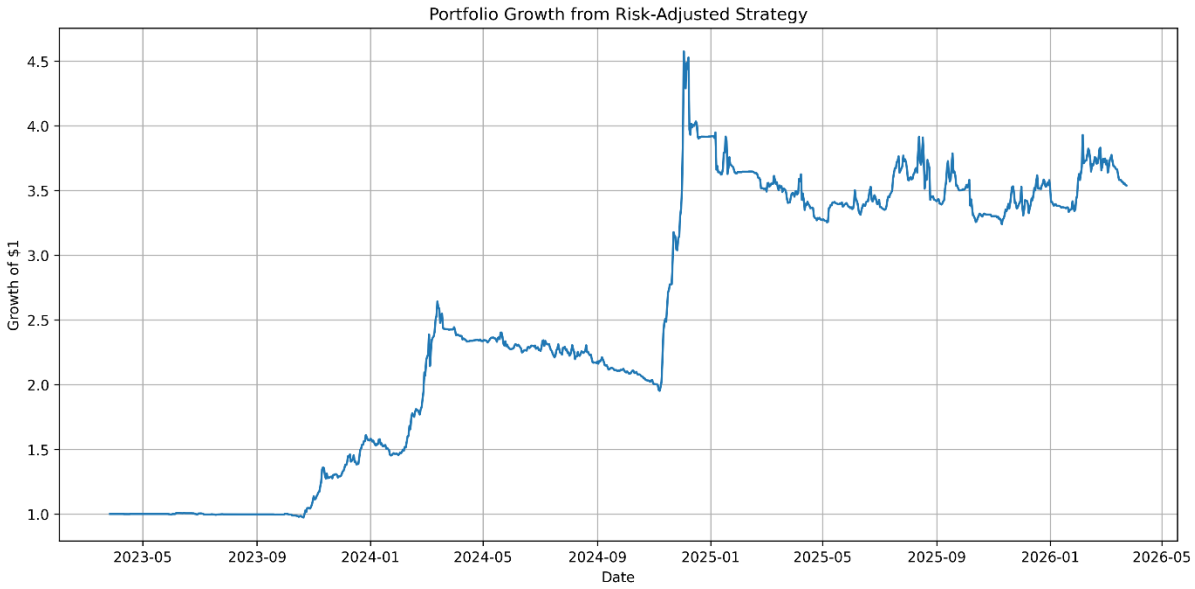

Figure 11.3.1.1: Portfolio Growth from Risk-Adjusted Strategy

Figure 11.3.1.1 shows how a $1 investment grows using the risk-adjusted Agentic AI strategy over the full backtesting period.

The portfolio stays mostly flat until mid-2023, indicating low activity or cautious trading. From September 2023, it grows rapidly, reaching about $2.65 in early 2024, aligning with the start of a broader crypto bull market.

A sharp peak of around $4.55 occurs in late 2024, likely due to a strong market rally. After that, the portfolio stabilizes between $3.25 and $4.00 during 2025 and early 2026.

Overall, the investment ends at about $3.54, resulting in a 253.8% total return.

11.3.2. Strategy vs Benchmark Comparison

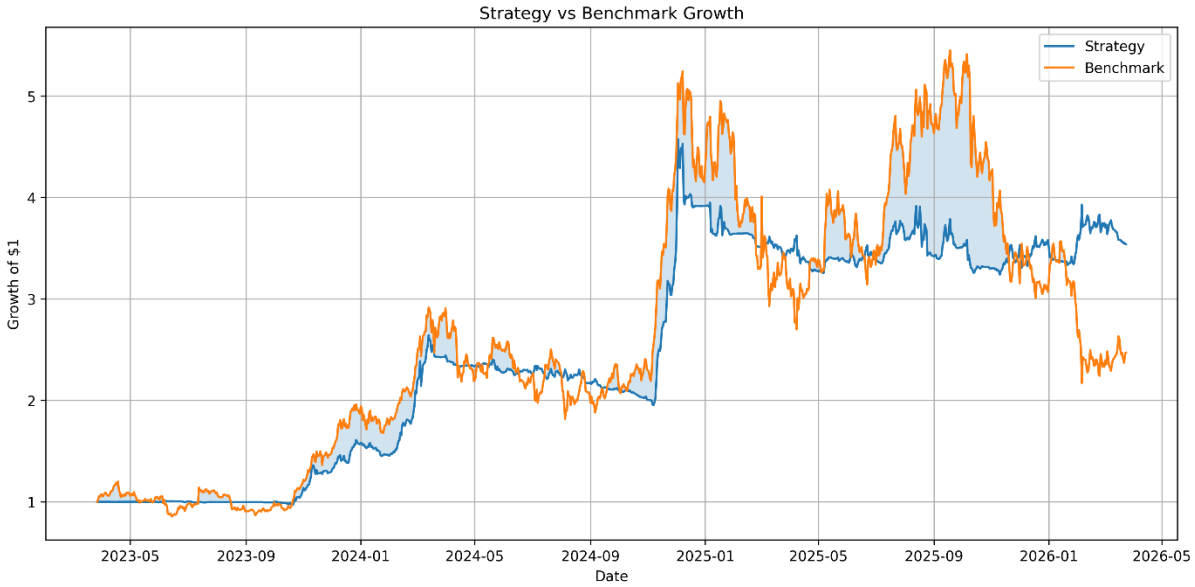

Figure 11.3.2.1: Strategy vs Benchmark Growth

Figure 11.3.2.1 compares the strategy (blue) with the benchmark (orange), where the shaded area shows when the strategy performs better.

Both follow a similar growth trend during 2023–2024, as the overall market rises. However, after late 2024, they start to differ. The benchmark peaks higher around 5.3x, but then drops sharply to about 2.4x by early 2026. In contrast, the strategy avoids large losses and finishes at 3.538x.

The main advantage of the strategy is better risk management, as it reduces losses during market downturns and keeps more value compared to the benchmark. So the strategy protects the traders from big losses.

11.3.3. Drawdown Analysis

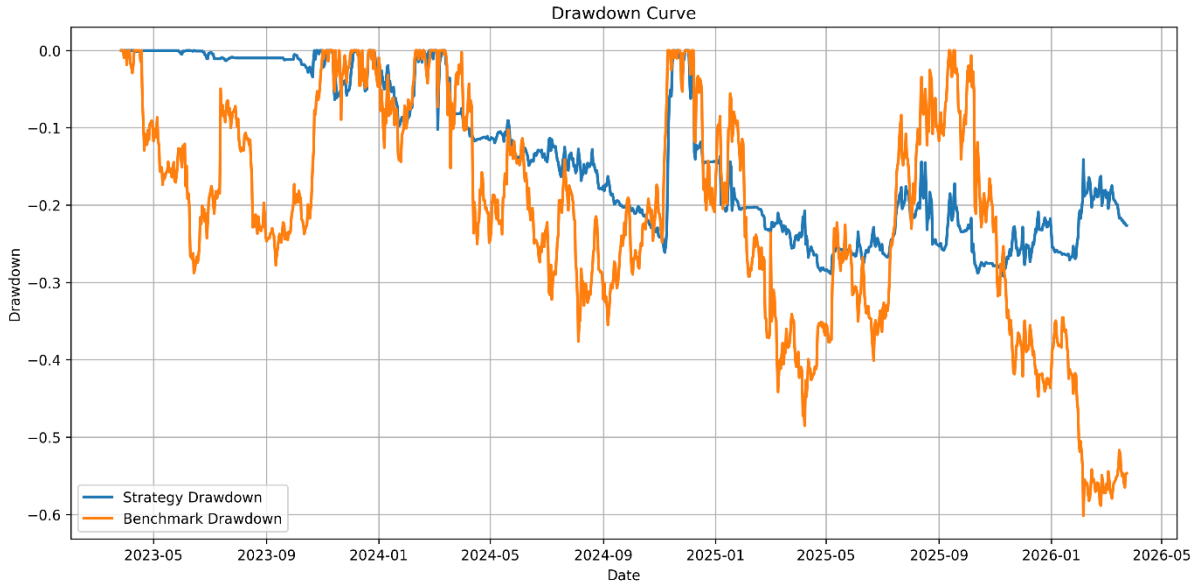

Figure 11.3.3.1: Strategy Drawdown vs Benchmark Drawdown

Figure 11.3.3.1 shows how much each strategy loses from its peak (drawdown).

- The Agentic AI strategy has a maximum loss of about -29%.
- The benchmark drops much more, around -60%, which is over twice as bad.

The benchmark experiences several big losses, including a major crash by early 2026. In contrast, the strategy keeps losses more controlled and recovers more steadily, with only one drop beyond -25%.

From 2025 onward, the benchmark keeps falling deeper, while the strategy stays more stable. This suggests the system reduced risk or avoided bad market conditions effectively.

11.3.4. Return Distribution Analysis

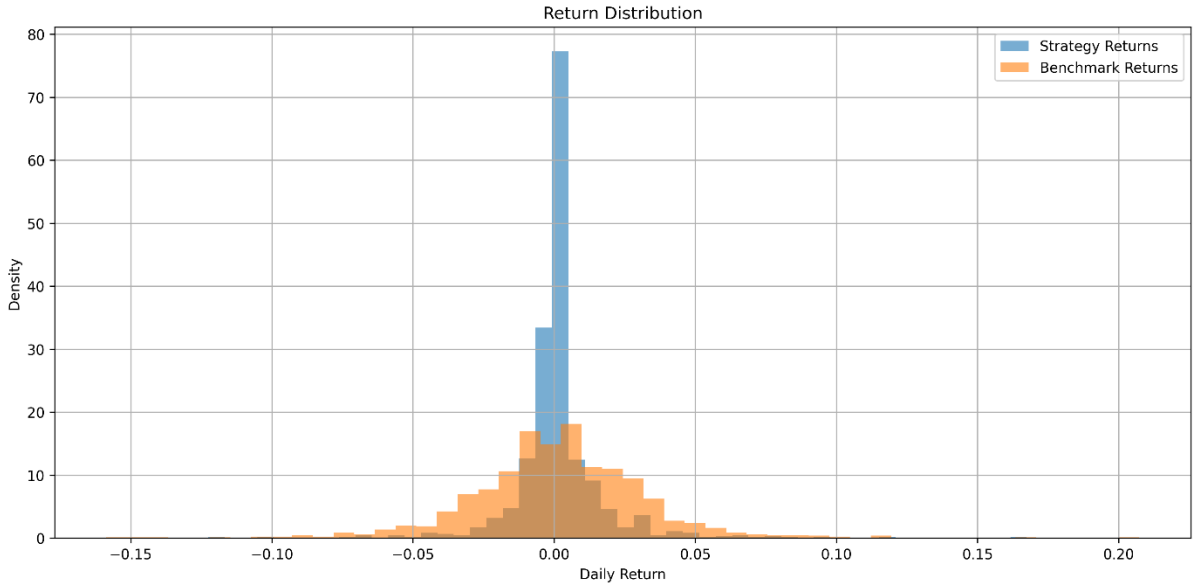

Figure 11.3.4.1: Return Distribution: Strategy Returns vs Benchmark Returns

Figure 11.3.3.1 shows how daily returns are distributed for both strategies.

- The Agentic AI strategy mostly stays close to 0% daily return, meaning it often takes small or no positions and avoids big market swings.
- The benchmark has a much wider range of returns large gains and large losses, showing high volatility.
- The benchmark can earn bigger profits on some days, but also suffers bigger losses.
- The strategy gives up some big gains but greatly reduces large losses.

The strategy focuses on controlling risk, keeping returns stable, and avoiding major losses rather than chasing extreme profits.

11.3.5. Win/Loss Analysis

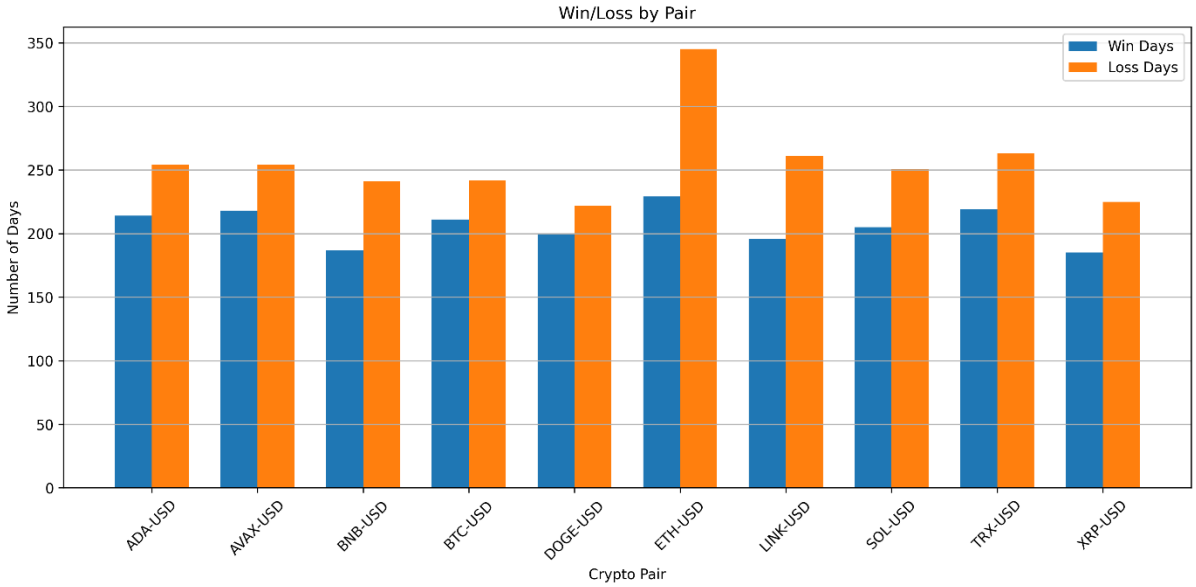

Figure 11.3.5.1: Win/Loss Day Count by Cryptocurrency Pair across the backtesting Period

Figure 11.3.5.1 shows how many days each asset made a profit (winning days) versus a loss (losing days).

- For all assets, losing days are more frequent than winning days which means win rate is below 50%.
- ETH has the most trading activity, with many winning and losing days.
- XRP and BNB have fewer winning days, suggesting weaker performance or less trading.

  Even with a 47.3% win rate, the strategy is still highly profitable.

- This means winning trades make more money than losing trades lose.
- The strategy cuts losses quickly and lets profits grow, which leads to strong overall returns.

11.3.6. Risk –Adjusted Outperformance

The key result is that the strategy has a much higher Sharpe Ratio value of 1.259 than the benchmark value of 0.667. This means the strategy gives better returns for the amount of risk taken. A Sharpe Ratio above 1 is considered strong, especially in volatile markets like crypto.

The strategy achieves this by earning higher total returns (Agentic AI strategy 253.8% vs. Benchmark 146.9%) and Keeping risk and volatility lower.

The system adjusts its positions based on market conditions, reducing exposure during bad periods instead of always staying invested.

11.3.7. Downside Protection

The comparison of maximum losses shows how well the strategy manages risk.

- The strategy loses about 29% at worst, while the benchmark loses about 60%.
- A 60% loss needs a 150% gain to recover, but a 29% loss needs only about 41%.

This shows that smaller losses are much easier to recover from.

The chart also shows that the strategy recovers faster after losses and avoids long and severe downturns seen in the benchmark. The strategy helps protect and grow wealth more effectively over time by controlling losses and reacting to bad market conditions.

12. Cloud Integration (AWS)

12.1. Architecture Overview

The system is implemented using a cloud-based architecture on Amazon Web Services (AWS). The architecture separates computation, storage, and structured data management to ensure scalability and efficiency.

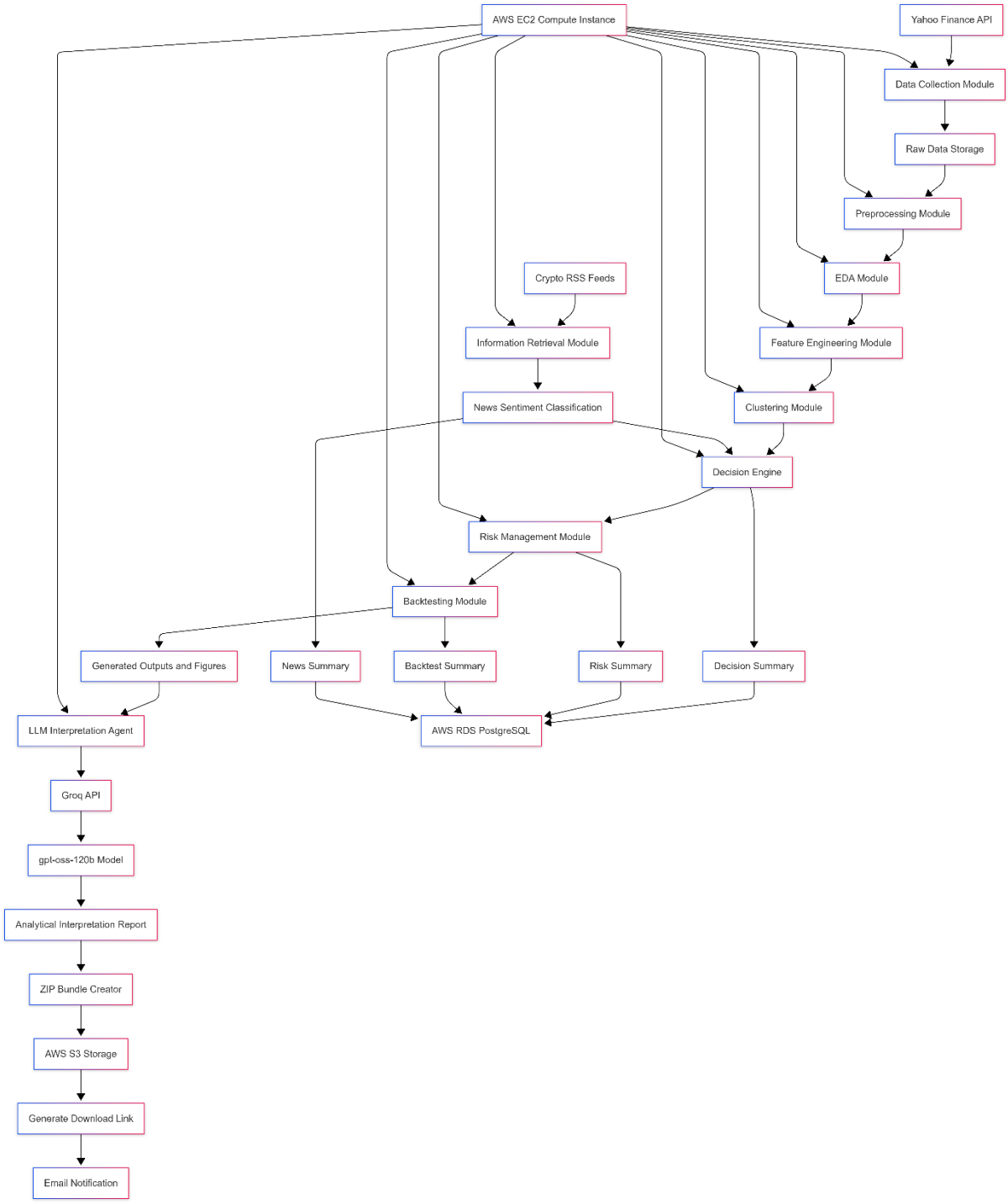

Figure 12.2.1: System Architecture

12.2. Amazon EC2 (Computation)

Amazon EC2 instance that running on Linux environment executes the entire workflow. The EC2 instance hosts the Python environment, dependencies, and project code. This allows the system to run end-to-end processing, including data collection, analysis, and model evaluation, in a centralized and scalable environment.

12.3. Amazon S3 (Storage)

The system uses Amazon S3 to store raw datasets, processed data, and generated outputs such as figures and reports. This provides a reliable and scalable storage solution that separates data from computation.

12.4. Amazon RDS PostgreSQL (Structured Storage)

Amazon RDS PostgreSQL database contains structured outputs, including news sentiment summaries, decision summaries, risk summaries, and backtesting results. This enables efficient querying and long-term storage of analytical results.

12.5. Execution Workflow

The Agentic AI system executes the complete workflow on EC2 using a single entry point script. After processing, the system uploads files to S3, and inserts structured results into RDS. This ensures a fully cloud-integrated pipeline.

13. Operational Improvement Recommendations

Operational improvements could enhance the system's performance and flexibility. Using advanced Natural Language Processing (NLP) models, such as transformer-based architectures, could make sentiment analysis more accurate. Reinforcement learning could enable adaptive trading strategies that adjust to changing market conditions. Integrating real-time data would allow live trading and continuous decision-making, while portfolio optimization could allocate capital more efficiently across assets. Finally, adding monitoring and alert systems would provide users with real-time insights and risk awareness.

14. Limitations

Several limitations affect the interpretation of the results. The system only considers the past data and simplifies considering Real-world factors such as transaction costs, slippage, and liquidity constraints mean the strategy may reduce actual performance. The strategy also has a win rate below 50% (47.3%), relying on larger gains from winning trades, which may be sensitive to changing market conditions. Additionally, the three-year testing period may not fully represent all possible market cycles, but here the system is believed to ignore less relevant data.

From a system perspective, the sentiment analysis is based on simple keyword matching, which may miss deeper context in news data, and the use of RSS feeds may limit information coverage. The decision engine uses fixed rules rather than adaptive learning, reducing flexibility in dynamic markets. Finally, the approach assumes that historical patterns will repeat, which may not always be true in real trading environments.

15. Conclusion

This paper presented an Agentic AI workflow for automated cryptocurrency trading analysis, integrating market analysis, information retrieval, risk management, and backtesting within a unified framework. The system supports more comprehensive trading decisions by combining technical indicators with external news data, while cloud integration enhances scalability.

Backtesting across a ten-asset portfolio from March 2023 to March 2026 demonstrates consistent outperformance over a passive equal-weight benchmark across multiple dimensions. Return distribution confirms that these gains stem from selective, risk-controlled positioning and asymmetric payoffs rather than aggressive exposure or high win rates.

These results support the applicability of Agentic AI frameworks in cryptocurrency portfolio management, particularly for downside risk mitigation in volatile markets. Future research should include out-of-sample testing, realistic transaction cost modelling, and validation across more market cycles and assets.

16. Question 02

16.1. Binomial Distribution

The Binomial distribution is a discrete probability distribution that models the number of successes in a fixed number of independent trials, where each trial has only two possible outcomes of success or failure. Each trial follows the same probability of success.

16.1.1. Mathematical Formula

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{(n-k)}$$

Where:

n = number of trials

k = number of successes

p = probability of success in one trial

(1 - p) = probability of failure

16.1.2. Key Characteristics

- ✓ Fixed number of trials (n is constant)
- ✓ Each trial is independent
- ✓ Only two outcomes per trial:
    - Success (1)
    - Failure (0)
- ✓ Probability of success (p) remains constant across trials

16.1.3. Mean and Variance

- ✓ Mean (Expected value)

  $\mu = np$

  When increase n, expected successes increase

- ✓ Variance

  $\sigma^2 = np(1 - p)$

  Variance depends on both of success and failure probabilities

16.1.4. Shape of Distribution

- If p = 0.5, it is symmetric distribution
- If p > 0.5, it is left-skewed
- If p < 0.5, it is right-skewed

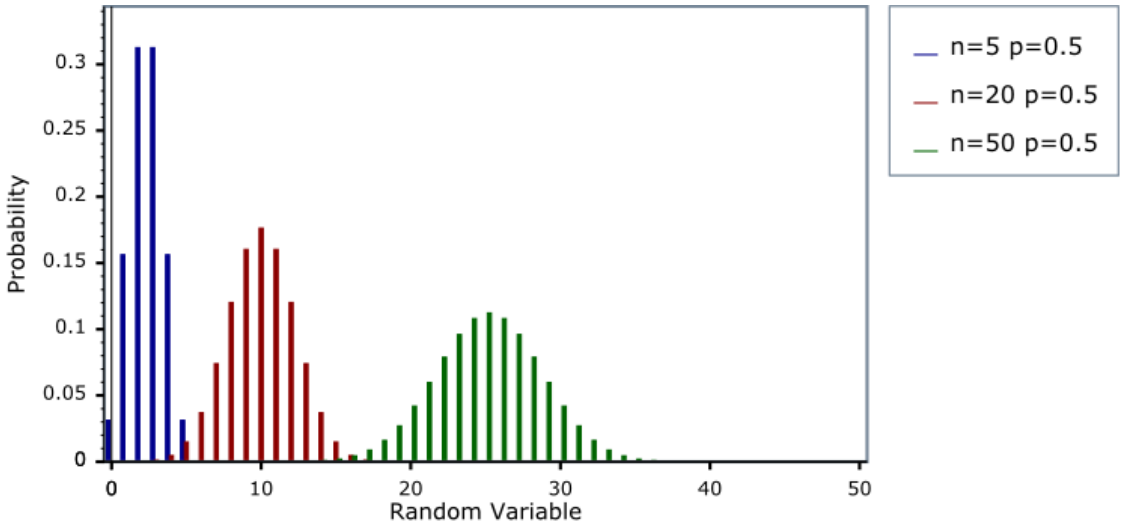

Figure 16.1.4.1: Binomial Distribution, Success fraction p is fixed at 0.5 and sample size n is vary

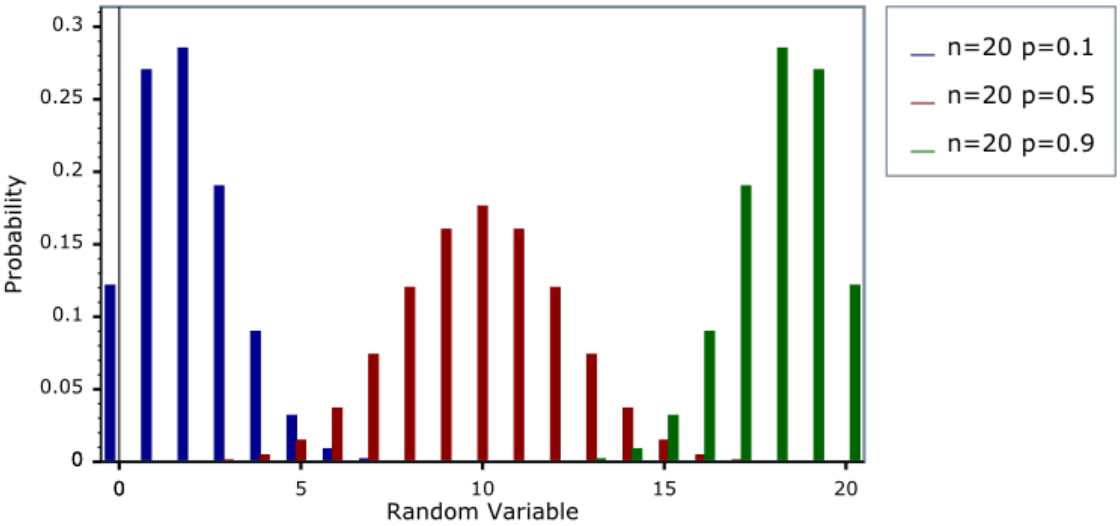

Figure 16.1.4.2: Binomial Distribution, Sample size n is fixed at 20 and success fraction p is vary

[3]

16.1.5. Example

A trader tests a strategy over 10 days, where each day success (profit) probability is 0.6 and failure (loss) probability is 0.4.

Let: X = number of profitable days

Then

$$X \sim Binomial(n = 10, p = 0.6)$$

1. Compute the probability of exactly 6 profitable days.

$$P(X = 6) = \binom{10}{6} 0.6^6 (0.4)^{(10-6)}$$

$$P(X = 6) \approx 0.251$$

2. Compute the probability of at least 8 profitable days.

$$P(X \geq 8) = \sum_{k=8}^{10} \binom{10}{k} 0.6^k (0.4)^{(10-k)}$$

$$P(X \geq 8) \approx 0.167$$

16.1.6. Real-World Applications

- Finance: number of profitable trades in a fixed period
- Quality control: number of defective items in a batch
- Healthcare: number of patients responding to treats
- Machine learning: binary classification outcomes

16.2. Poisson Distribution

The Poisson distribution is a discrete probability distribution that models the number of times an event occurs in a fixed interval of time, space, or volume, given that;

- Events occur independently
- The average rate (λ) is constant
- Two events cannot occur at exactly the same instant

16.2.1. Mathematical Formula

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Where:

λ = number of occurrences in the interval

k = number of events

e ≈ 2.718

16.2.2. Key Characteristics

- ✓ Counts number of events in an interval
- ✓ Events occur randomly and independently
- ✓ The average rate (λ) is constant

16.2.3. Means and Variance

- ✓ Mean (Expected value)

  $\mu = \lambda$

- ✓ Variance

  $\sigma^2 = \lambda$

  In Poisson Distribution Mean = Variance which is a unique property.

16.2.4. Shape of Distribution

- When λ is small, it is highly right-skewed
- When λ is increased, it becomes more symmetric
- For large λ, it approximates normal distribution

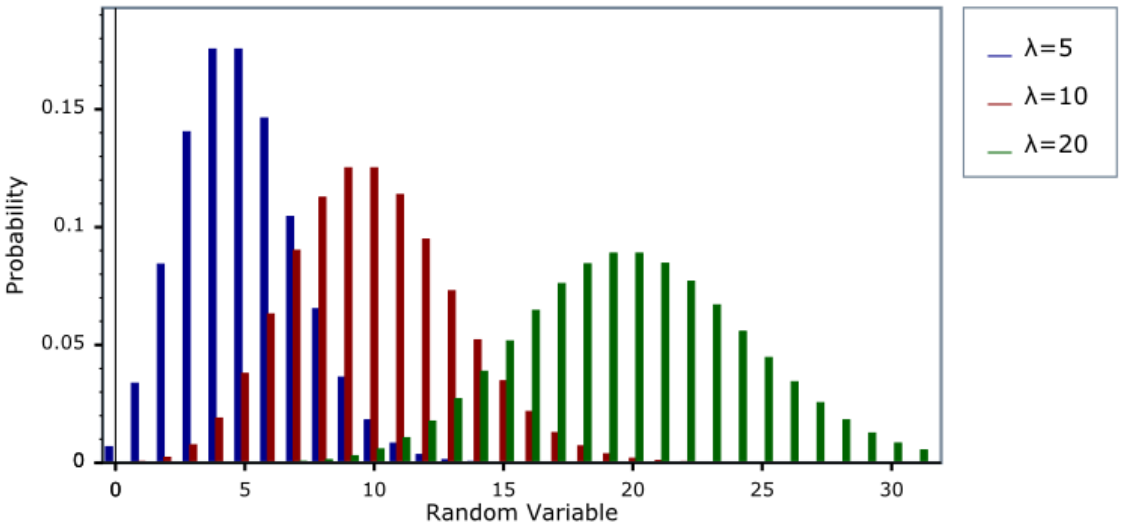

Figure 16.2.4.1: Poisson Distribution for various λ values

[4]

16.2.5. Relationship with Binomial Distribution

Poisson is a limiting incident of binomial distribution when n is very large and p is very small, np equals to λ.

16.2.6. Example

Suppose a cryptocurrency change receives an average of 5 large buy orders per hour.

Let: X = number of buy orders per hour

Then

$$X \sim Poisson(\lambda = 5)$$

1. Calculate the probability of exactly 3 large orders per hour

$$P(X = 3) = \frac{5^3 e^{-5}}{3!}$$

$$P(X = 3) \approx 0.14$$

2. Calculate the probability of no large orders

$$P(X = 0) = \frac{5^0 e^{-5}}{0!}$$

$$P(X = 0) \approx 0.0067$$

3. Calculate the probability of more than 8 orders

$$P(X > 8) = 1 - \sum_{k=0}^{8} \frac{5^k e^{-5}}{k!}$$

$$P(X \leq 8) \approx 0.9319$$

$$P(X > 8) = 1 - 0.9319 = 0.0681$$

16.2.7. Real-World Applications

- Finance:
    - Number of trades occurring per minute
    - Sudden spikes in trading activity
- Telecommunications: number of calls arriving in a system
- Healthcare: number of patients arriving at a hospital
- Web systems: number of requests per second

16.3. Geometric Distribution

The Geometric distribution is a discrete probability distribution that models the number of trials required to achieve the first success in a sequence of independent Bernoulli trials. Each trial has two outcomes (success/failure) and a constant probability of success (p).

16.3.1. Mathematical Formula

$$P(X = k) = p(1 - p)^{k-1}$$

Where:

k = number of trials until the first success

p = probability of success

(1 – p) = probability of failure

16.3.2. Key Characteristics

- ✓ Models waiting time until first success
- ✓ Trials are independent
- ✓ Probability of success (p) is constant
- ✓ The number of trials is not fixed

16.3.3. Means and Variance

- ✓ Mean (Expected value)

$$\mu = \frac{1}{p}$$

When p goes to smaller, it make longer expected waiting time and larger p makes quicker success

- ✓ Variance

$$\sigma^2 = \frac{1 - p}{p^2}$$

16.3.4. Memoryless Property

$$\mathrm{P}(\mathrm{X} > \mathrm{s} + \mathrm{t} \mid \mathrm{X} > \mathrm{s}) = \mathrm{P}(\mathrm{X} > \mathrm{t})$$

If there have already been s trials without success, the probability of needing more t trials is the same as starting freshly. The probability of the next success does not depend on past failures because the process “resets” after each trial.

16.3.5. Shape of Distribution

- Always right-skewed
- High probability at small values of k
- Probability decreases exponentially when k increases

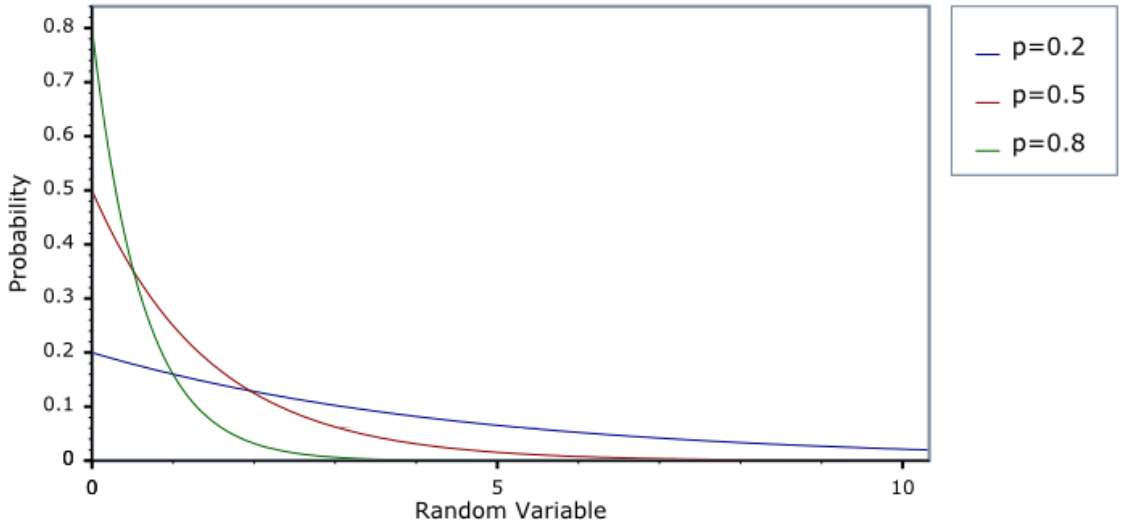

Figure 16.3.5.1: Geometric Distribution for various success fraction p values

[5]

16.3.6. Example

A trading strategy has 47% chance of profitable trade.

Let: X = number of trades until the first profitable trade

Then

$$X \sim Geometry(\mathrm{p} = 0.47)$$

1. Calculate the probability of first success occurs on 3rd trade

$$P(X = 3) = 0.47(1 - 0.47)^{3-1}$$

$$P(X = 3) \approx 0.132$$

16.3.7. Real-World Applications

- Finance: Number of trades until first profitable trade
- Quality Testing: Trials until first defect is found
- Marketing: Attempts until first customer conversion
- Systems: Retries until first successful connection

16.4. Hypergeometric Distribution

The Hypergeometric distribution is a discrete probability distribution that models the number of successes in a sample drawn without replacement from a finite population. Unlike the Binomial Distribution, the probability changes after each draw because items are not replaced.

16.4.1. Mathematical Formula

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

Where:

N = total population size

K = number of successes in population

n = sample size

k = number of observed successes

16.4.2. Key Characteristics

- ✓ Sampling happens without replacement
- ✓ Probability of success changes each draw
- ✓ Population size is finite
- ✓ Trials are dependent

16.4.3. Means and Variance

- ✓ Mean (Expected value)

$$\mu = \mathrm{n}\frac{K}{N}$$

- ✓ Variance

$$\sigma^2 = \mathrm{n}\left(\frac{K}{N}\right)\left(\frac{N-K}{N}\right)\left(\frac{N-n}{N-1}\right)$$

16.4.4. Shape of Distribution

- Can be symmetric or skewed depending on parameters
- Similar to Binomial when population is large
- Becomes narrower due to reduce randomness

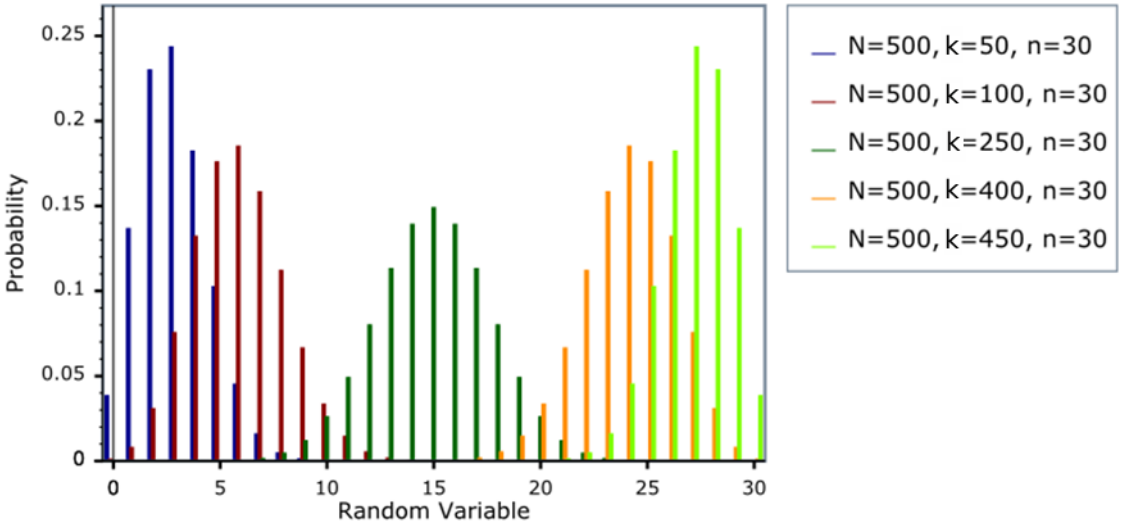

Figure 16.4.4.1: Hypergeometric Distribution: The proportion of success k changes, while keeping population and sample size

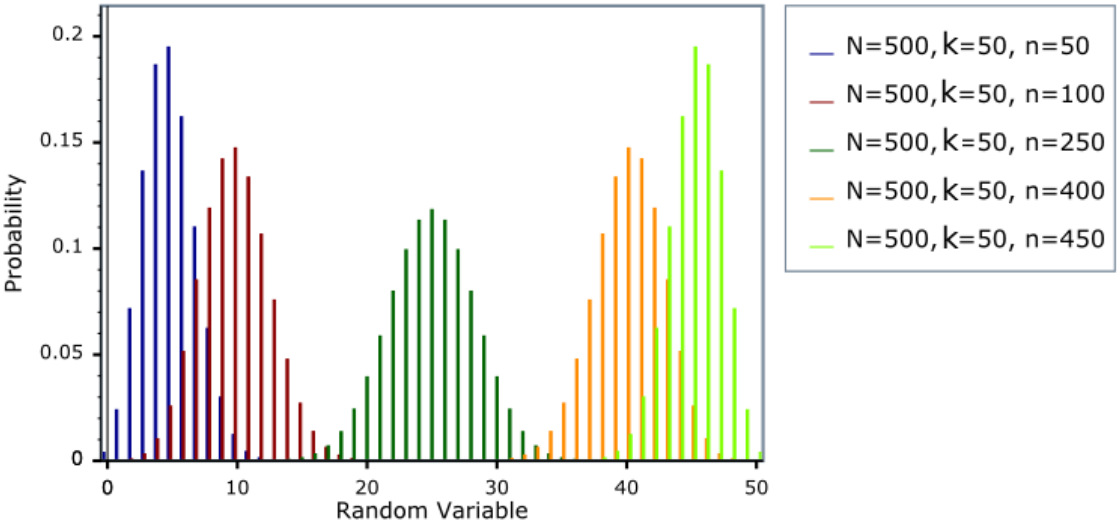

Figure 16.4.4.2: Hypergeometric Distribution: The sample size changes, while keeping population and the proportion of success.

Figure 16.4.5.1 and Figure 16.4.5.2 shows same shape of graph

[6]

16.4.5. Example

A trader analyze a portfolio of 10 cryptocurrencies where, 4 are profitable and 6 are not. The trader randomly select 5 cryptocurrencies without replacement.

Let: X = number of profitable cryptocurrencies

Then

$$X \sim Hypergeometry(\mathrm{N} = 10, \mathrm{K} = 4, \mathrm{n} = 5)$$

1. Calculate the probability of selecting exactly 2 profitable cryptocurrencies

$$P(X = 2) = \frac{\binom{4}{2}\binom{10-4}{5-2}}{\binom{10}{5}}$$

$$P(X = 2) \approx 0.476$$

2. Calculate the probability of selecting at least 3 profitable cryptocurrencies

$$P(X \geq 3) = \sum_{k=3}^{5} \frac{\binom{4}{k}\binom{10-4}{5-k}}{\binom{10}{5}}$$

$$P(X \geq 3) \approx 0.262$$

16.4.6. Real-World Applications

- Finance: Selecting assets from a fixed portfolio
- Quality control: Sampling defective items without replacement
- Card games: Drawing specific cards from a deck
- Auditing: Sampling records from a dataset

16.5. Negative Binomial Distribution

The Negative Binomial distribution is a discrete probability distribution that models the number of trials required to achieve a fixed number of successes (r^{th} Success) in a sequence of independent Bernoulli trials.

16.5.1. Mathematical Formula

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

Where:

k = total number of trials

r = required number of successes

p = probability of success

(1−p) = probability of failure

16.5.2. Key Characteristics

- ✓ Models number of trials until r successes
- ✓ Trials are independent

- ✓ Probability of success (p) is constant

16.5.3. Means and Variance

- ✓ Mean (Expected value)

$$\mu = \frac{r}{p}$$

- ✓ Variance

$$\sigma^2 = \frac{r(1-p)}{p^2}$$

Expected trials increase when r increases and higher variance compare to Binomial shows more spread.

16.5.4. Shape of Distribution

- Typically right-skewed
- Becomes more symmetric when r increases
- Spread depends heavily on p

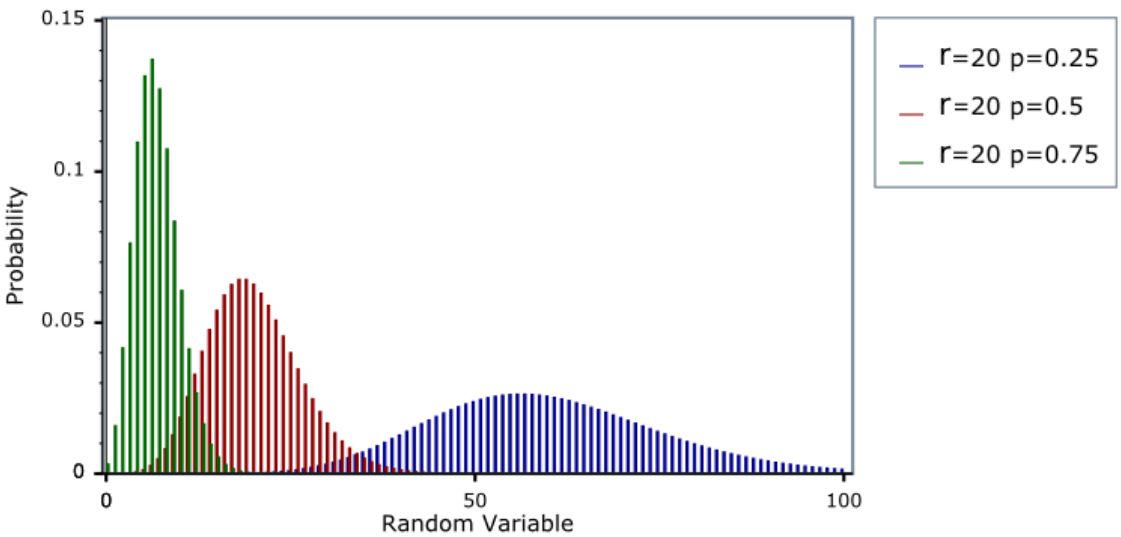

Figure 16.5.4.1: Negative Binomial Distribution: The success fraction changes while keeping number of success trials same

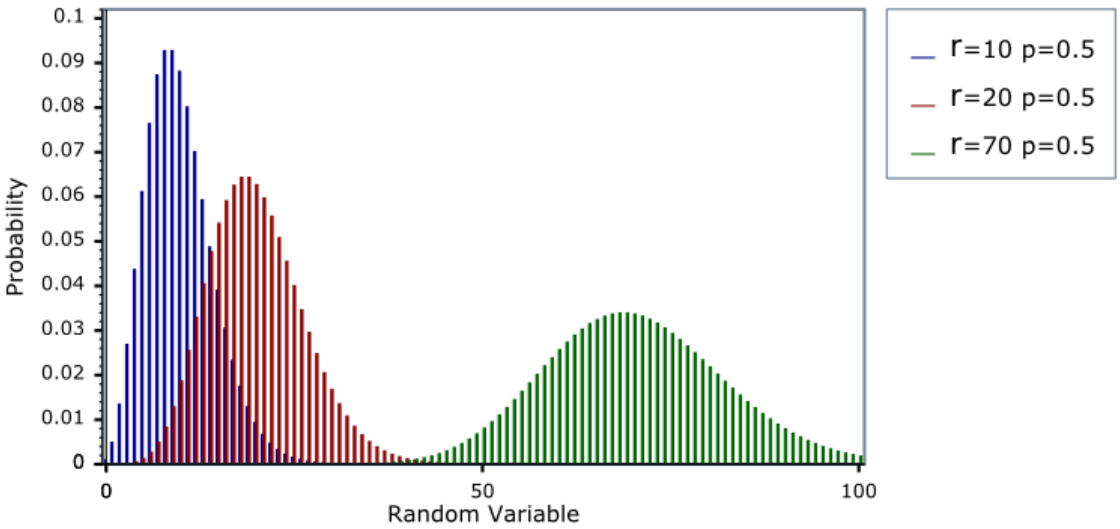

Figure 16.5.4.1: Negative Binomial Distribution: The number of success trials changes while keeping success fraction p same

[7]

16.5.5. Relationship with Geometric Distribution

Geometric is a special case of the Negative Binomial when r equals to 1, Geometric and Negative Binomial become same.

16.5.6. Example

A trading strategy has 47% chance of profitable trade. Need to know, how many trades are needed to achieve 3 profitable trades.

Let: X = number of trades until r^{th} success trade

Then

$$X \sim Negative\ Binomial(\mathrm{r} = 3, \mathrm{p} = 0.47)$$

1. Compute the probability of the 3rd success occurs on 8th trade

$$P(X = 8) = \binom{8-1}{3-1} 0.47^3 (1 - 0.47)^{8-3}$$

$$P(X = 8) \approx 0.089$$

   There is an 8.9% probability to occur the 3rd success on 8th trade.

2. Find the expected number of trades needed

$$E(X) = \frac{3}{0.47}$$

$$E(X) \approx 6.38$$

16.5.7. Real-World Applications

- Finance: Number of trades needed to reach a target profit count
- Insurance: Number of claims until a certain number of events occur
- Marketing: Attempts until multiple conversions
- Healthcare: Trials until a number of recoveries

16.6. Normal Distribution

The Normal distribution is a continuous probability distribution that describes data that is symmetrically distributed around the mean, forming **a** bell-shaped curve. This is the widely used statistical probability distribution because many real-world variables naturally follow this.

16.6.1. Mathematical Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

μ = mean (center of distribution)

σ = standard deviation (spread)

σ^2 = variance

16.6.2. Key Characteristics

- ✓ Symmetrical around the mean
- ✓ Mean = Median = Mode
- ✓ Defined by two parameters of mean (μ) and standard deviation (σ)
- ✓ Curve is bell-shaped

16.6.3. Shape of Distribution

- Peak at the mean
- Tails extend infinitely
- Standard Deviation controls the spread

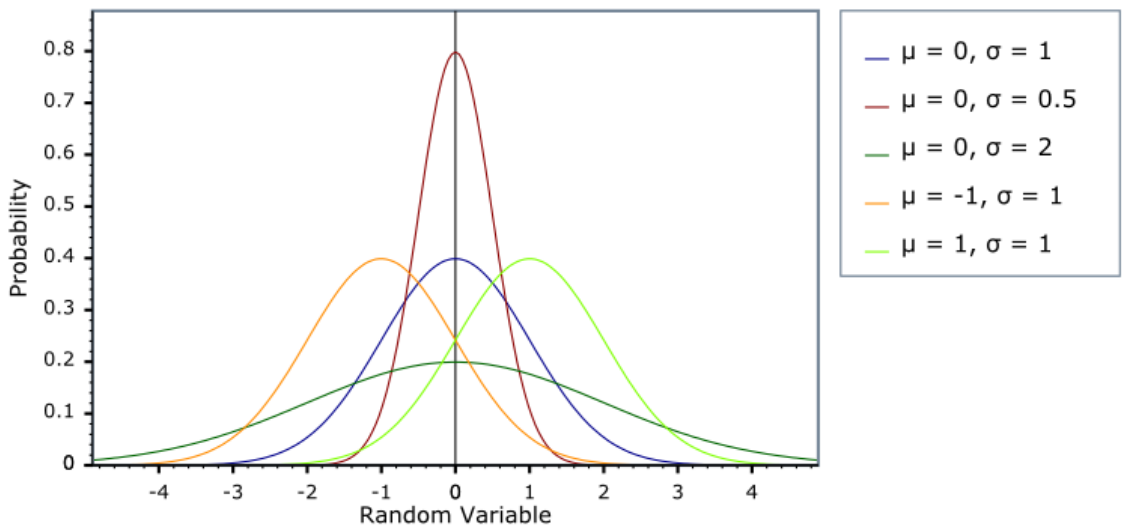

Figure 16.6.3.1: Normal Distribution: Various mean values and standard deviation values

[8]

The empirical rule is 68% of data lies within ±σ, 95% within ±2σ, and 99.7% within ±3σ.

16.6.4. Relationship with Other Distribution

When n is large and p not close to 1 or 0, Normal Distribution approximates Binomial Distribution.

$$Binomial(n, p) \approx Normal(np, np(1-p))$$

When λ is large, Normal Distribution approximates Poisson Distribution.

$$Poisson(\lambda, \lambda) \approx Normal(\lambda, \lambda)$$

16.6.5. Example

A cryptocurrency gives the mean value of 0.001176 and standard deviation value of 0.0246.

Let:

$$X \sim Normal(\mu = 0.001176, \sigma = 0.0246)$$

1. Calculate the probability of extreme losses.

   Typically, the returns below -2σ from the mean defines "Extreme Losses".

   Threshold = μ - 2σ = 0.001176 – 2(0.0246) = -0.048024

$$Standardize\ (Z - score) = \frac{X - \mu}{\sigma}$$

$$Z = \frac{-0.048024 - 0.001176}{0.0246} = -2$$

From standard normal table

$$P(Z < -2) \approx 0.02275$$

$$P(extreme\ loss) \approx 0.0228$$

There is about 2.28% chance of extreme loss.

2. Calculate the expected range of returns.

   Using the 68, 95, 99.7 rule,

   1 standard deviation (Typical Range)

$$\mu \pm \sigma = 0.001176 \pm 0.0246$$

$$[-0.0234, 0.0246]$$

   2 standard deviation (Wider Range)

$$\mu \pm 2\sigma = 0.001176 \pm 2(0.0246)$$

$$[-0.0480, 0.0504]$$

   3 standard deviation (Extreme Range)

$$\mu \pm \sigma = 0.001176 \pm 3(0.0246)$$

$$[-0.0726, 0.0740]$$

   68% of returns: $[-0.0234, 0.0246]$

   95% of returns: $[-0.0480, 0.0504]$

   99.7% of returns: $[-0.0726, 0.0740]$

3. Find risk levels using standard deviation.

- Low Risk Zone (±1σ)

  Most common outcomes

  Small gains/losses

  Range: −2.34% to +2.58%

- Moderate Risk (±2σ)

  Occasional large swings

  Range: −4.80% to +5.04%

- High Risk / Extreme Events (±3σ)

  Rare but significant moves

  Range: −7.26% to +7.40%

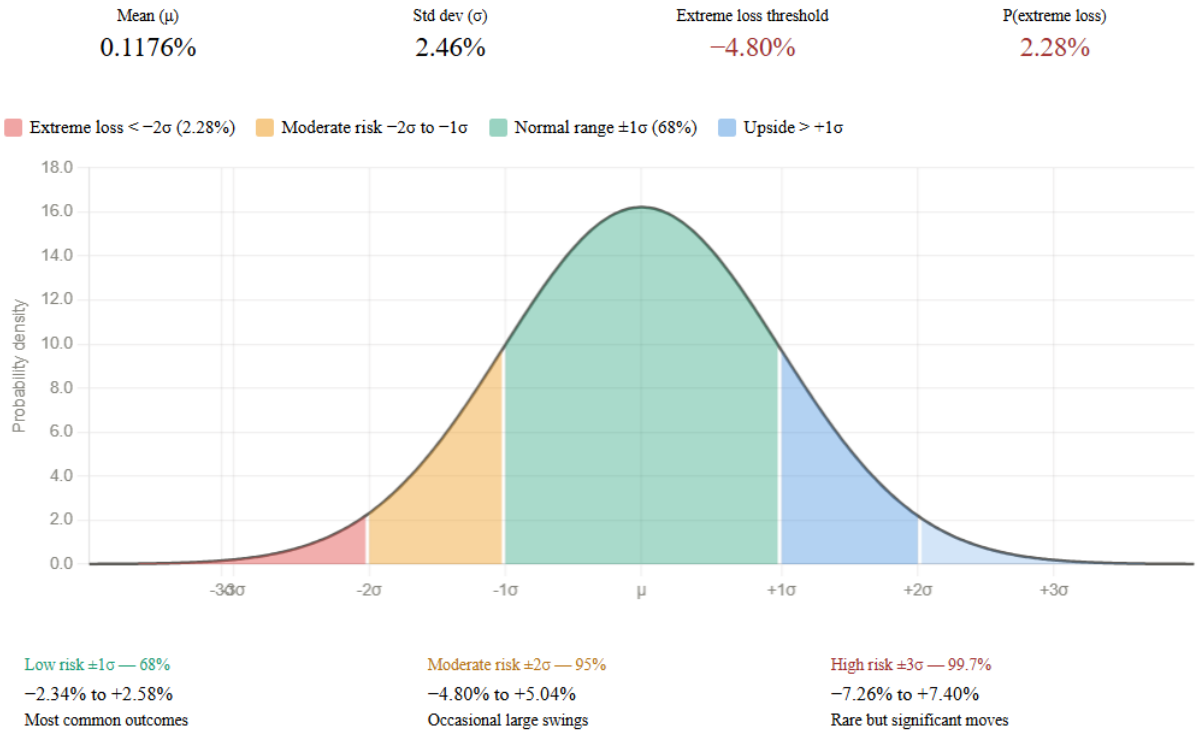

Figure 16.6.5.1: Expected ranges and risk levels

16.6.6. Real-World Applications

- Finance:
  - Stock/crypto return modelling
  - Risk analysis (Value at Risk)
- Machine learning: Feature distributions
- Natural sciences: Measurement errors
- Quality control: Product variations

16.7. Exponential Distribution

The Exponential distribution is a continuous probability distribution that models the time between consecutive events in a process where events occur independently at a constant average rate.

16.7.1. Mathematical Formula

$$f(x) = \lambda e^{-\lambda x}, \qquad x \geq 0$$

Where:

λ = rate of occurrence (events per unit time)

x = time between events

16.7.2. Key Characteristics

- ✓ Models waiting time between events
- ✓ Events occur independently
- ✓ Constant rate (λ)
- ✓ Defined only for x ≥ 0

16.7.3. Means and Variance

- ✓ Mean (Expected value)

$$\mu = \frac{1}{\lambda}$$

- ✓ Variance

$$\sigma^2 = \frac{1}{\lambda^2}$$

When larger λ, it is shorter waiting time.

16.7.4. Memoryless Property

$$P(X > s + t \mid X > s) = P(X > t)$$

If there have already been s time wait until an event occurs, the probability of needing more t time is the same as starting freshly. The process "forgets" that waiting time. The future waiting time behaves exactly like a brand-new start.

16.7.5. Shape of Distribution

- Always right-skewed
- Starts high at x = 0
- Decreases exponentially
- Has a long tail

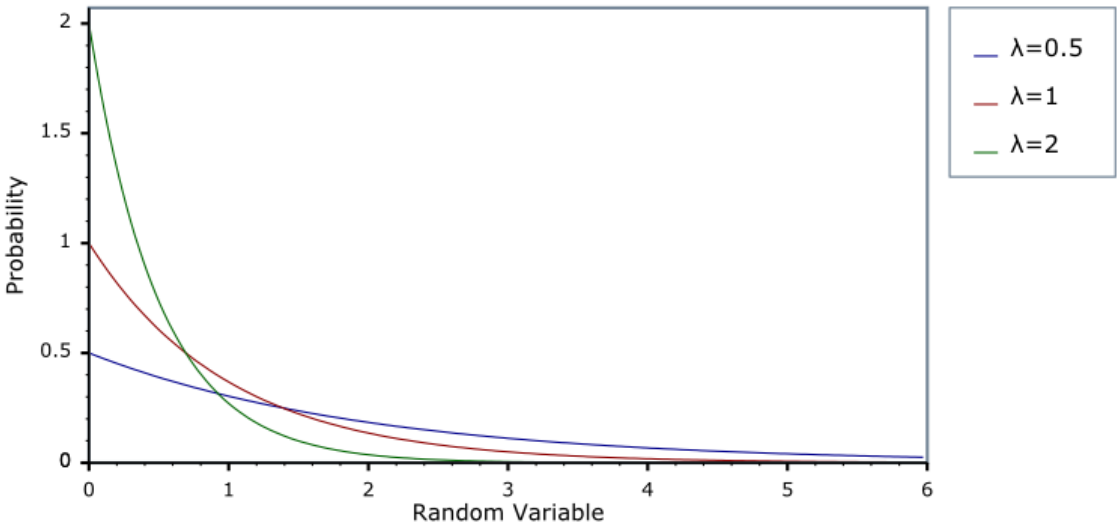

Figure 16.7.5.1: Exponential Distribution for different values of lambda

[9]

16.7.6. Relationship with Poisson Distribution

Poisson Distribution consider the number of events in fix interval of time, but Exponential Distribution consider the time between two events.

16.7.7. Example

Suppose a cryptocurrency exchange receives 10 large trades per hour on average

Let:

$$X \sim Exponential(\lambda = 10)$$

1. Calculate the probability the next trade occurs within 5 minutes

Converts time to hour

$$5\ minutes = \frac{5}{60} = \frac{1}{12} hours$$

Using CDF

$$P(X \leq t) = \int_0^t \lambda e^{-\lambda x}\, dx = 1 - e^{-\lambda t}$$

$$P(X \leq 5\ min.) = 1 - e^{-10\frac{1}{12}} = 1 - e^{-0.8333}$$

$$P(X \leq 5\ min.) = 1 - 0.4346 = 0.5654$$

There is about a 56.54% probability that the next large trade occurs within 5 minutes.

2. Expected waiting time

$$\frac{1}{10} = 0.1\ hours = 6\ minutes$$

16.7.8. Real-World Applications

- Finance: Time between trades
- Telecommunications: Time between incoming calls
- Reliability engineering: Time until system failure
- Web systems: Time between user requests

16.8. Chi-Squared Distribution

The Chi-Squared distribution is a continuous probability distribution that arises as the sum of the squares of independent standard normal variables. Analyzers widely uses this in hypothesis testing, goodness-of-fit tests, and independence tests.

16.8.1. Mathematical Formula

$$x^2 = \sum_{i-1}^{k} Z_i^2$$

Where:

$Z_i \sim N(0,1)$ Standard normal variables

$$Z = \frac{Observed\ value - Expected\ value}{Expected\ value}$$

k = degrees of freedom

16.8.2. Key Characteristics

- ✓ Values are always non-negative (≥ 0)
- ✓ Shape is right-skewed, especially for small k
- ✓ Shape depends on degrees of freedom

16.8.3. Means and Variance

- ✓ Mean (Expected value)

  $\mu = \mathrm{k}$

- ✓ Variance

  $\sigma^2 = 2k$

When k increase, distribution becomes more symmetric and for large k, Chi-Squared Distribution approximates Normal Distribution

16.8.4. Shape of Distribution

- When small k, shape is highly right-skewed
- When large k, shape is more bell-shaped

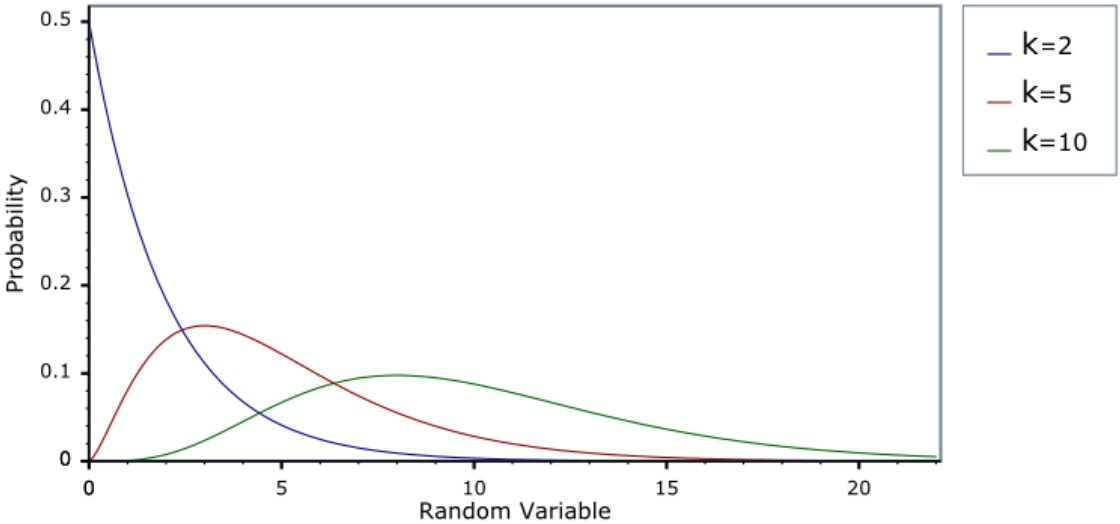

Figure 16.8.4.1: Chi-Squared for various degrees of freedom (k) values

[10]

16.8.5. Example

Suppose the Agentic AI system gives the summary of Sell, Buy, and Hold.

Signal	Observed (O)	Expected (E)
Buy	2064	3643.33
Sell	2558	3643.33
Hold	6308	3643.33

Interpret the signal accuracy

Calculate each component

$$x^2 = \sum_{i-1}^{k} \frac{(O-E)^2}{E_i}$$

Buy

$$\frac{(2064 - 3643.33)^2}{3643.33} \approx 684.6$$

Sell

$$\frac{(2558 - 3643.33)^2}{3643.33} \approx 323.3$$

Hold

$$\frac{(6308 - 3643.33)^2}{3643.33} \approx 1948.9$$

Total X^2

$$X^2 = 684.6 + 323.3 + 1948.9 \approx 2956.8$$

Degrees of freedom (df)

$$df = k - 1 = 3 - 1 = 2$$

At 5% significance level

$$X^2_{0.05,2} \approx 5.99$$

Final Decision

$$2956.8 \gg 5.99$$

Signal distribution is not consistent with expected distribution

16.8.6. Real-World Applications

- Statistics:
  - Goodness-of-fit tests
  - Test of independence (e.g., contingency tables)
- Finance: Model validation
- Machine learning: Feature selection (Chi-square test)
- Quality control: Checking expected vs actual outcomes

16.9. Extended Comparison

Distribution	Type	Focus	Key Parameter(s)	Special Feature
Binomial	Discrete	Success count	n, p	Fixed trials
Poisson	Discrete	Event count	λ	Mean = Variance
Geometric	Discrete	First success timing	p	Memoryless

Hypergeometric	Discrete	Sampling without replacement	N, K, n	Dependent trials
Negative Binomial	Discrete	r-th success timing	r, p	Flexible success count
Normal	Continuous	Continuous data	μ, σ	Bell curve
Exponential	Continuous	Time between events	λ	Memoryless
Chi-Squared	Continuous	Variance / deviation	k	Based on squared normals

17. References

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Appendix A: Data and Outputs

The `agentic_ai_trading_comscds252p008/docs/sample_data_outputs.zip` file that is stores in the repository, contains all the sample data and output files.

A.1: Data

A.1.1: Raw Data (data/raw)

Historical cryptocurrency data retrieved from Yahoo Finance. `Combined_crpto_data.csv` file contains merged each cryptocurrency data.

- ✓ data/raw/ada_usd.csv
- ✓ data/raw/avax_usd.csv
- ✓ data/raw/bnb_usd.csv
- ✓ data/raw/btc_usd.csv
- ✓ data/raw/combined_crpto_data.csv
- ✓ data/raw/doge_usd.csv
- ✓ data/raw/eth_usd.csv
- ✓ data/raw/link_usd.csv
- ✓ data/raw/sol_usd.csv
- ✓ data/raw/trx_usd.csv
- ✓ data/raw/xrp_usd.csv

A.1.2: Processed Data (data/processed)

A.1.2.1: Cleaned Data

data/processed/cleaned_crypto_data.csv

A.1.2.2: Clustered Data

data/processed/clustered_crypto_data.csv

A.1.2.3: EDA Enriched Data

data/processed/eda_enriched_crypto_data.csv

A.1.2.4: Featured Data

data/processed/featured_crypto_data.csv

A.1.2.5: Signal Summary

data/processed/signal_summary.csv

A.1.3: Output Data (data/outputs)

This folder contains results generated by workflow

A.1.3.1: Crypto News Headlines

data/outputs/crypto_news_headlines.csv

A.1.3.2: Crypto News Sentiment Summary

data/outputs/crypto_news_sentiment_summary.csv

A.1.3.3: Market Data with News Signals

data/outputs/market_data_with_news_signal.csv

A.1.3.4: Market Data with Decisions

data/outputs/market_data_with_decisions.csv

A.1.3.5: Decision Summary

data/outputs/decision_summary.csv

A.1.3.6: Backtest Results

data/outputs/backtest_results.csv

A.2: Reports

This folder contains all visualizations and summary tables used in the final analysis.

A.2.1: Figures (reports/figures)

A.2.1.1: Interactive Correlation Heatmap

reports/figures/interactive_correlation_heatmap.html

A.2.1.2: Interactive Daily Returns

reports/figures/interactive_daily_returns.html

A.2.1.3: Interactive Normalized Price Trends

reports/figures/interactive_normalized_price_trends.html

A.2.1.4: Interactive Rolling Volatility

reports/figures/interactive_rolling_volatility.html

A.2.2: Tables (reports/tables)

A.2.2.1: Descriptive Statistics

reports/tables/descriptive_statistics.csv

A.2.2.2: Asset Clusters

reports/tables/asset_clusters.csv

A.2.2.3: Cluster Summary

reports/tables/cluster_summary.csv

A.2.2.4: Correlation Matrix

reports/tables/correlation_matirx.csv

A.2.2.5: Backtest Summary

reports/tables/backtest_summary.csv